

Experiment 13:

Implement the Car Price Prediction Model using Python

Python Code:

```
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([
    [2015, 50000],
    [2016, 45000],
    [2017, 40000],
    [2018, 35000],
    [2019, 30000],
    [2020, 25000],
    [2021, 20000],
    [2022, 15000],
    [2023, 10000],
    [2024, 5000]
])

y = np.array([450000, 480000, 520000, 560000, 600000,
              650000, 700000, 750000, 800000, 850000])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

new_car = np.array([[2022, 18000]])

predicted_price = model.predict(new_car)

print("Actual prices:", y_test)
```

```
print("Predicted prices:", np.round(y_pred, 2))
print("Mean Squared Error:", round(mean_squared_error(y_test, y_pred), 2))
print("R2 Score:", round(r2_score(y_test, y_pred), 2))
print("New car details:", new_car[0])
print("Predicted car price:", round(predicted_price[0], 2))
```

Output:

```
Actual prices: [800000 480000]
Predicted prices: [792500. 477500.]
Mean Squared Error: 31250000.0
R2 Score: 1.0
New car details: [ 2022 18000]
Predicted car price: 720500.0
```